Subrata Samanta

Ph.D. Student (PMRF Fellow), Department of Physics,
Indian Institute of Technology, Kanpur 208 016, India



Specialization:	**Theoretical High Energy Physics**
Ph.D. Supervisor:	**Prof. Debtosh Chowdhury** Associate Professor, Indian Institute of Technology, Kanpur, India

General Information

Place and Date of Birth:	Purba Medinipur, 721 634, India | August, 1999
Address:	Department of Physics, Indian Institute of Technology, Kanpur 208 016, India
Phone:	+91-7980402785 (cell)
email:	samantaphy20@iitk.ac.in | samantaphys@gmail.com
url:	home.iitk.ac.in/~samantaphy20/ (Homepage)
	inspirehep.net/authors/2685933/ (INSPIRE)
	github.com/samantaphys/ (GitHub)

Research Interests

Theoretical High-Energy Physics	Particle Physics Beyond the Standard Model and Higgs Physics, Precision and Flavor Physics, Collider Physics.

Education

Aug 2022 - Present	Ph.D. in Physics, Indian Institute of Technology, Kanpur, CPI: **9.5**
Sep 2020 - May 2022	M.Sc. in Physics, Indian Institute of Technology, Kanpur, CPI: **8.9**
Aug 2017 - Oct 2020	B.Sc. in Physics, University of Calcutta, Percentage: **88.8%** **First class - First position** in College (Scottish Church College, Kolkata, India)

Dissertation / Thesis Work

Aug 2022 - *Ongoing* *Ph.D. Thesis*	Thesis Title : **Higgs as a probe of physics beyond the Standard Model** Indian Institute of Technology, Kanpur, India Supervisor: Prof. Debtosh Chowdhury [debtoshc@iitk.ac.in]
May 2021 - Apr 2022 *Master's Thesis*	Thesis Title : **Pressure and Shear distributions inside the proton** Indian Institute of Technology, Kanpur, India Supervisor: Prof. Dipankar Chakrabarti [dipankar@iitk.ac.in]

Achievements and Awards

Aug 2022 - Present	**Prime Minister's Research Fellowship**, MHRD, Government of India (Direct Entry, Indian Institute of Technology, Kanpur, India)
Mar 2023	Awarded **Dhirendra Mohan Saha Roy Gold Medal** (Scottish Church College, Kolkata, India)
Mar 2023	Awarded **The Certificate of Merit**, Scottish Church College, Kolkata, India (For academic excellence in The Bachelor of Science)
Feb 2021	Awarded **Late Srikant Mishra Scholarship** (Indian Institute of Technology, Kanpur, India)
Oct 2019	National Topper of the **National Graduate Physics Examination** (Selected for Winter Camp in Physics at BITS- Pilani, Goa, India)

Long Stays

Nov 4-21, 2024 *Europe tour*	Uppsala University, Sweden, Nov 4-8, 2024 Bonn University, Germany, Nov 11-15, 2024 Karlsruhe Institute of Technology (KIT), Germany, Nov 18-21, 2024

Internships

May 2022 - Jul 2022 *Summer Intern*	Research Title : **Anisotropic SU(3) Gauge Fields in Lattice QCD** National Institute of Science Education and Research, Bhubaneswar, India Supervisor: Prof. Subhasish Basak [sbasak@niser.ac.in]

Workshops and Schools

Jul 1-13, 2024 *School-Cum-Workshop*	**Machine Learning for Particle and Astroparticle Physics (ML4HEP)** Institute of Physics, Bhubaneswar, India
Jun 16-27, 2025 *Summer School*	**Summer School on Particle Physics (SMR 4080)** Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy
Jan 18-30, 2026 *Winter School*	**School for Advanced Topics in Particle Physics (SATPP)** International Centre for Theoretical Science (ICTS), Bengaluru, India

Conference Talks

Oct 14-18, 2024	**17th International Conference on Interconnections between Particle Physics and Cosmology (PPC) 2024**, Hyderabad, India
Nov 4-8, 2024	**HIGGS 2024**, Uppsala, Sweden

Invited Talks

Nov 11, 2024	**Bethe Center for Theoretical Physics (BCTP)**, Bonn University, Bonn, Germany
Nov 19, 2024	**Institute for Theoretical Particle Physics (TTP)**, KIT, Karlsruhe, Germany
Feb 3, 2026	**Centre for High Energy Physics (CHEP)**, IISc, Bengaluru, India

Teaching Experience

2025 - Present *Online*	Tutor at **Bankura Christian College, West Bengal, India** Subject: Physics
2023 - Present *Online*	Mentor at **NPTEL, Ministry of Education, India** Subject: Physics Through Computational Thinking & Thermal Physics
2022 - Present *Offline*	Teaching Assistance at **Indian Institute of Technology, Kanpur, India** Subject: Phyics (UG & PG) Laboratory, Particle Physics
2023 - 2024 *Online & Offline*	Tutor at **Raghunathbari Ramtarak High School, West Bengal, India** Subject: Physics
2023 - 2024 *Online & Offline*	Tutor at **Deriachak Sri Aurobindo Vidyamath (H.S.), West Bengal, India** Subject: Physics
2023 - 2024 *Online*	Mentor at **SATHEE, Ministry of Education, India** Subject: Physics
Spring, 2024 *Online*	Offered a course on "**Mathematical Methods in Physics - I**" PMRF lecture series, Institute of Smart Structures and Systems (ISSS), India

References

Prof. Debtosh Chowdhury
Associate Professor (Ph.D. Supervisor),
Department of Physics,
Indian Institute of Technology (IIT) Kanpur,
Kanpur 208 016, India,
Tel: +91-512-259-2151,
debtoshc@iitk.ac.in

Prof. Kirtimaan Mohan
Assistant Professor (Collaborator),
Department of Physics,
Michigan State University,
East Lansing, MI 48824-1046, US,
Tel: +91-7760106923,
kamohan@msu.edu

Prof. Anirban Kundu
Professor (Collaborator),
Department of Physics, University of Calcutta,
Kolkata 700009, India,
Tel: +91-9830619645,
akphy@caluniv.ac.in

Prof. Dipankar Chakrabarti
Professor (M.Sc. Supervisor),
Department of Physics, IIT Kanpur
Kanpur 208 016, India,
Tel: +91-512-259-6696,
dipankar@iitk.ac.in

Entrance Exam Qualification

Mar 2022	Joint Entrance Screening Test in Physics, India (for PhD)
Jun 2021	Joint CSIR - UGC NET Examination in Physics, India (for Junior Research Fellowship)
Apr 2020	Joint Admission test for Master's in Physics, India (for MSc)
Mar 2020	Joint Entrance Screening Test in Physics, India (for Integrated PhD)
Mar 2019	National Graduate Physics Examination, India (National Topper)

Course-work at IIT Kanpur

Spring 2023	PHY780A: High Energy Physics II PHY680: Particle Physics
Fall 2022	PHY627A: Computer Simulation Methods in Physics PHY690M: Advanced General Relativity & Black Holes
Spring 2022	PHY680A: Particle Physics PHY524A: Atomic, Molecular, and Optical Physics
Fall 2021	PHY461A: Experimental Physics I PHY681A: Quantum Field Theory PHY407A: Special and General Relativity
Summer 2021	PHY441A: Electronics
Spring 2021	PHY412A: Statistical Mechanics PHY432A: Quantum Mechanics II PHY422A: Mathematical Methods II PHY552A: Classical Electrodynamics I PHY526A: Nuclear and Particle Physics
Fall 2020	PHY401A: Classical Mechanics I PHY431A: Quantum Mechanics I PHY473A: Computational Physics PHY421A: Mathematical Methods I PHY667A: High Energy Astrophysics of Binary Star Systems

Computer Skills

Operating systems:	MacOS, Linux/Unix, Windows
Programming languages:	Python, C++, C, Fortran
Calculus languages:	Mathematica
Documentation and Visualization:	LaTeX, Wiki, Inkscape, Matplotlib
Web development:	HTML, CSS, JavaScript, Responsive Design

Numerical Skills

Open-source code	**HEPfit** : Markov Chain Monte Carlo package based on Bayesian Statistics **Bertini** : Package for Solving Systems of Polynomial Equations
Mathematica tools	**FeynRules, FeynArts, FormCalc, FeynCalc, LoopTools**

Languages

Mother Tongue:	Bengali
Good:	English, Hindi

Publications

1. D. Chowdhury, A. Kundu, P. Mondal, **S. Samanta**, and A. Srivastava, *One-loop renormalization and ρ parameter in the Georgi-Machacek model*, [2601.06252].**
2. P. Mondal and **S. Samanta**, *Light Scalars in the Extended Georgi-Machacek Model*, [2506.06427].**
3. D. Chowdhury, P. Mondal, and **S. Samanta**, *Next-to-Leading Order Unitarity Fits in the Extended Georgi-Machacek Model*, [2404.18996].**
4. **S. Samanta**, A. Acharya, J. Mitra, and S. Bhattacharyya, *Saha-Basu equation of state and its application to Carnot cycle*, *Eur. J. Phys.* **43** (2022) 045101.

**Note: Authors are listed in alphabetical order.*